Finite projections of SYK conformal dynamics and near-horizon scattering

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Finite projections of the SYK conformal sector retain different aspects of holographic dynamics according to their spectral resolution. Using the conformal operator dynamics of the Sachdev–Ye–Kitaev model and the JT eikonal interaction, we relate a finite Krylov chain to positive scattering channels through a common representation of the conformal algebra. We derive an exact boundary-feedback equation: occupation of the last Krylov site supplies the term that redirects the continuum operator growth. The gravitational response is controlled instead by the low-product momentum weight selected by the increasing horizon boost. Return probabilities resolve interference between momentum products, while entanglement resolves products of momentum differences. Its zero-frequency weight explains how the deterministic interaction realizes the phase-ensemble average purity set by channel participation. Equal-resource comparisons expose the resulting observable dependence: preserving many exchange moments, resolving soft momenta, and distributing quantum coherence impose distinct requirements on a finite model. These mechanisms connect operator spreading, gravitational scattering, and register entanglement within an explicit quantum Hamiltonian and give physical criteria for choosing its resolution.

I. INTRODUCTION

The spreading of a local perturbation connects strongly interacting matter to the redistribution of quantum information. Holography relates such boundary dynamics to gravitational processes [1]; near a black-hole horizon, a growing relative boost amplifies the interaction of weak probes [2], with the thermal chaos bound constraining the growth rate [3]. Jackiw–Teitelboim (JT) gravity [4, 5] isolates the reparametrization sector and its eikonal scattering response [6]. The Sachdev–Ye–Kitaev (SYK) model supplies a microscopic many-body setting for this infrared structure [7–9]. Krylov representations organize operator spreading into a chain generated by repeated commutators [10], while dynamical symmetry and orthogonal polynomials make the conformal evolution explicit [11, 12]. The full operator-complexity trajectory encodes the Lanczos sequence [13]; spectral support, probe choice, and the dynamical regime shape its growth and its relation to chaos [14–17]. These connections make finite collective representations a natural setting in which to examine how a quantum model retains the physical content of a holographic response.

Recent progress has made the bulk interpretation increasingly concrete. Double-scaled SYK chord states provide a bulk Hilbert space [18], and their Krylov dynamics gives geometric length observables [19, 20], with extensions to finite-temperature sine-dilaton gravity [21]. Matter insertions, higher moments, and shockwaves broaden this dictionary [22–24]; exact symmetry-resolved Krylov correlators and the near-horizon interpretation of the symmetry-generated chain further connect Krylov evolution to particle motion [25, 26]. On the many-body side, binary sparse couplings and random Pauli-spin interactions simplify microscopic realizations [27, 28], while sparsity-dependent tests identify regimes with preserved Lyapunov growth [29]. Noisy quantum simulations resolve return probabilities and out-of-time-order correlators [30], and circuit-complexity geometry gives a complementary interpretation of Krylov distance [31]. Open-system studies relate dissipation to operator growth, edge modes, and relaxation in SYK [32–35]. Scramblon approaches resum nonlinear chaotic response and its fluctuations [36–38]; random-spin signatures [39], the error-resilient reversal demonstrated by Li *et al.* [40], and multiround echo protocols [41] connect this structure to controllable measurements. Information-recovery studies further distinguish different diagnostics of scrambling [42]. However, constructing a finite quantum representation of these processes requires deciding which spectral features must be resolved for each observable and how this choice changes its dynamics.

Finite Hamiltonian representations have a well-developed foundation in orthogonal-polynomial bath discretization and dynamical error bounds [43, 44], with Gaussian quadrature supplying their spectral nodes and weights [45]. Krylov truncation errors admit a chain-reflection and Loschmidt-echo interpretation [46]; open chain boundaries, localization, and transport-based break times give complementary accounts of the accessible evolution interval [47–49]. Conditional-phase interactions organize entanglement through products of spectral differences [50], while fixed-amplitude phase ensembles determine its average through participation weights [51]. Measurement-adapted reduction places the observable's resolution at the center of model selection [52]. In this paper, we apply these ideas to a common conformal representation of SYK operator dynamics and the positive momentum measure of JT scattering. We evaluate the finite-chain acceleration identity as an exact terminal-occupation feedback term and connect the projected parabolic generator to finite scattering channels. Equal-channel comparisons then identify the distinct resolution requirements of the regulated four-point response, return probability, and register entanglement: low momentum products govern the boost response, while conditional-phase differences govern the redistribution of coherence. We determine how the deterministic interaction realizes the phase-ensemble purity average and use reversal ob-

servables to examine its sensitivity to coherent errors and dephasing. The resulting spectral criteria specify which collective dynamics a finite quantum model retains.

II. SYK OPERATOR DYNAMICS AND THE CONFORMAL CHAIN

We begin with the Majorana SYK Hamiltonian

$$H_{\mathrm{SYK}} = \sum_{i<j<k<l} J_{ijkl}\chi_i\chi_j\chi_k\chi_l, \qquad \overline{J_{ijkl}^2} = \frac{3!J^2}{N^3}, \quad (1)$$

with $\{\chi_i, \chi_j\} = \delta_{ij}$ and independent, zero-mean Gaussian couplings. The thermal inner product $(A|B)_\beta = \mathrm{tr}(\rho_\beta^{1/2}A^\dagger\rho_\beta^{1/2}B)$ turns repeated commutators with H_{SYK} into an orthogonal Krylov chain. Taking the large-N, strong-coupling limit gives the conformal chain. A low-energy probe of scaling dimension Δ has the normalized thermal Wightman function and Lanczos coefficients [8, 10, 11]

$$C_{\mathrm{c}}(t) = \mathrm{sech}^r(\alpha t), \qquad r = 2\Delta, \qquad \alpha = \pi/\beta, \quad (2)$$

$$b_n = \alpha\sqrt{n(n+r-1)}. \quad (3)$$

For the single Majorana probe in Eq. (1), $\Delta = 1/4$, so $r = 1/2$. The basis label counts successive directions in operator space. The probability carried by each basis vector measures how the evolving probe spreads through this collective space.

The conformal coefficients form a positive discrete-series representation of $\mathfrak{sl}(2,\mathbb{R})$ [10, 11, 26]. Its generators act as

$$K_0|n\rangle = (n + r/2)|n\rangle, \quad (4)$$

$$K_+|n\rangle = \sqrt{(n+1)(n+r)}|n+1\rangle, \qquad K_- = K_+^\dagger, \quad (5)$$

and the chain Liouvillian is $\mathcal{L} = \alpha(K_+ + K_-)$. The mean Krylov position $\mathcal{K}(t) = \sum_n n|\varphi_n(t)|^2$ grows as $r\sinh^2(\alpha t)$. The positive parabolic generator

$$P = 2K_0 - K_+ - K_- \quad (6)$$

has a Gamma spectral measure in $|0\rangle$, equivalently obtained from the parabolic limit of symmetry-generated evolution [25]. A fixed conformal rotation relates it to the null generators whose boost weights determine longitudinal momentum scaling. This identifies the operator-space origin of the momentum measure used below. Appendix A gives the rotation and spectral construction.

Projecting onto $|0\rangle, \ldots, |m-1\rangle$ produces a finite chain. The projection modifies the conformal commutator at its last site, and the mean position obeys

$$\frac{\mathrm{d}^2\mathcal{K}_m}{\mathrm{d}(\kappa t)^2} = \mathcal{K}_m + \frac{r}{2} - \mathcal{B}_m(t), \qquad \kappa = 2\alpha, \quad (7)$$

$$\mathcal{B}_m(t) = \frac{m(m+r-1)}{2}|\varphi_{m-1}(t)|^2. \quad (8)$$

This specializes the Lanczos acceleration identity [12] to the projected conformal chain. The boundary occupation supplies the feedback that changes the outward acceleration, giving a closed expression for the chain-end effect underlying Krylov truncation errors [46, 47]. As the operator distribution reaches the chain end, the missing outgoing link redirects its motion and produces the finite-chain return structure. The microscopic thermal-inner-product construction and direct SYK checks are described in Appendix G.

III. HOLOGRAPHIC RESPONSE AND FINITE SPECTRAL PROJECTION

The JT eikonal interaction couples the longitudinal momenta of two probes [6]. For equal external dimensions and equal thermal regulation, its normalized four-point response is

$$F(z) = \int_0^\infty \mathrm{d}\mu_r(p)\int_0^\infty \mathrm{d}\mu_r(q)\, e^{-zpq}, \quad (9)$$

$$\mathrm{d}\mu_r(p) = \frac{p^{r-1}e^{-p}}{\Gamma(r)}\,\mathrm{d}p, \qquad z = z_0e^{\kappa t}. \quad (10)$$

The limit holds the ratio of relative boost to Schwarzian coupling fixed as both become large. Schwarzian bootstrap methods provide an exact description of this collective gravitational sector [53]. The conformal representation specifies each probe's momentum measure; the eikonal interaction specifies the gravitational coupling between the probes.

The projected operator P_m has diagonal entries $2n + r$ and nearest-neighbor matrix elements $-\sqrt{(n+1)(n+r)}$. Its eigenvalues p_j and vacuum spectral weights w_j are precisely the nodes and normalized weights of generalized Gauss–Laguerre quadrature. Diagonalizing this collective chain therefore gives the finite momentum channels. In the momentum basis, the quantum realization is

$$P_m = \sum_j p_j|j\rangle\langle j|, \qquad |v_m\rangle = \sum_j \sqrt{w_j}|j\rangle, \quad (11)$$

$$H_m = -P_m \otimes P_m, \qquad |\psi_m\rangle = |v_m\rangle \otimes |v_m\rangle. \quad (12)$$

The corresponding overlap and its regulated continuation are

$$A_m(s) = \langle\psi_m|e^{-\mathrm{i}sH_m}|\psi_m\rangle, \quad (13)$$

$$F_m(z) = A_m(\mathrm{i}z) = \sum_{j,k} w_jw_k e^{-zp_jp_k}. \quad (14)$$

Here s is the unitary scattering parameter, while z is the positive boost variable of the regulated response. The boundary time enters through the exponential relation in Eq. (10).

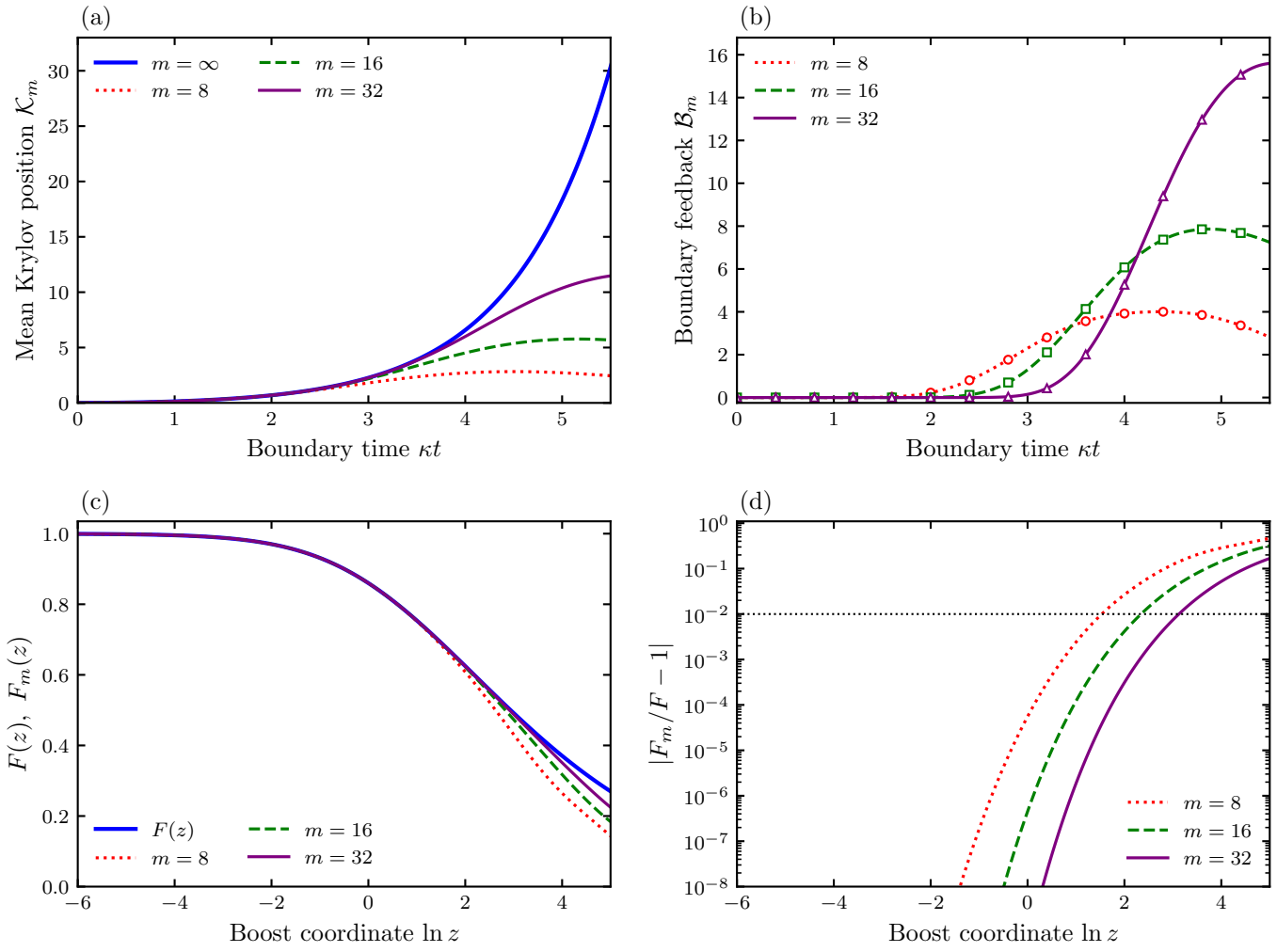


FIG. 1. Finite conformal SYK dynamics and the associated JT projection at $r = 1/2$. (a) Mean Krylov position for $m = 8, 16, 32$, compared with the infinite-chain law. (b) Boundary feedback $\mathcal{B}_m$; symbols evaluate $\mathcal{K}_m + r/2 - \mathrm{d}^2\mathcal{K}_m/\mathrm{d}(\kappa t)^2$ independently from the double commutator. (c) Continuous JT response and Gauss–Laguerre channel responses. (d) Relative response errors; the dotted horizontal line marks 1%. Panels (a),(b) use real boundary time, and panels (c),(d) use the regulated boost coordinate $\ln z$.

Figure 1 connects two manifestations of the projection. In operator space, the wavefunction encounters a terminal link and its growth acquires the boundary-feedback term. In momentum space, the same projection replaces a continuous positive generator by a discrete spectrum. The horizon boost then progressively selects momenta near its soft endpoint. These probes share a representation and a channel cutoff, while their evolution selects different parts of that representation.

The Gauss projection preserves the exchange moments

$$\sum_{j,k} w_j w_k (p_j p_k)^n = (r)_n^2, \qquad 0 \le n \le 2m-1, \tag{15}$$

and its regulated error satisfies

$$0 \le F(z) - F_m(z) \le c_m z^{2m}, \qquad z \ge 0. \tag{16}$$

The coefficient is sharp at the origin; its derivation is in Appendix C. This hierarchy controls the response to weak scattering. To examine the subsequent dynamics, we also construct positive measures from equal-probability bins and uniform momentum bins, using each bin's conditional mean as its node. The GL model is the conformal-chain projection; the bin models give alternative spectral allocations of the target measure. All three constructions use the same number of quantum channels. Their preparation weights and spectral resolution provide a direct test of what moment preservation retains at finite evolution time.

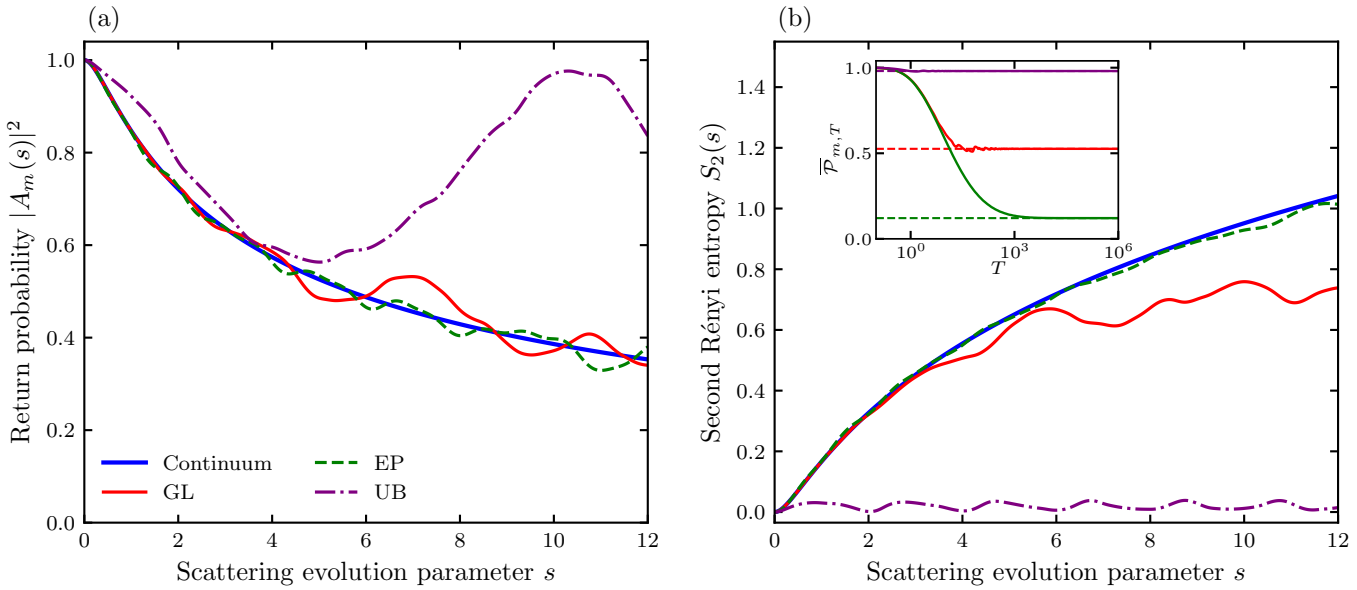


FIG. 2. Equal-resource quantum dynamics at $r = 1/2$, $m = 16$. (a) Return probability and (b) second Rényi entropy for the Gauss–Laguerre (GL), equal-probability (EP), and uniform-bin (UB) measures, compared with the continuum. The two bin constructions use the common omitted Gamma mass 10^{-12}. The inset in (b) shows the running time-averaged purity; dashed lines give $2I_2 - I_2^2$, with $I_2 = \sum_j w_j^2$. Every curve uses the same Hamiltonian construction and its own preparation weights. The inset averages purity before taking any logarithm.

IV. INTERFERENCE, ENTANGLEMENT, AND CHANNEL PARTICIPATION

Under H_m, each pair of momentum channels acquires a conditional phase. The return probability depends on interference among the products $p_j p_k$. As in product-interaction entanglement dynamics [50], the reduced-state purity resolves products of spectral differences, here sampled from the projected momentum measure:

$$\mathcal{P}_m(s) = \operatorname{tr} \rho_P(s)^2 = \mathbb{E}_m e^{\mathrm{i}s(p-p')(q-q')}, \tag{17}$$

$$S_{2,m}(s) = -\ln \mathcal{P}_m(s). \tag{18}$$

The four variables in this expectation are independent draws from the prepared channel measure. This expression identifies the spectral origin of entanglement: a partial trace removes phases common to different amplitudes and retains their differences.

The comparison in Fig. 2 exposes the distinction between the initial response and the later redistribution of coherence. Moment matching closely preserves the onset of evolution. Equal-probability sampling spreads the initial state over all channels, supporting a lower long-time average of purity. Uniform bins assign much of the probability to the first momentum interval for the singular endpoint measure used here, concentrating the state on a smaller effective set of channels. The resulting return and entropy profiles follow from these different distributions of phases and weights.

For distinct nodes, the difference spectrum contains a zero-frequency contribution whenever $p = p'$ or $q = q'$. Its weight fixes the exact infinite-time average

$$\overline{\mathcal{P}_m} = 2I_2 - I_2^2, \qquad I_2 = \sum_j w_j^2 = \frac{1}{N_{\mathrm{eff}}}. \tag{19}$$

This value coincides with the phase-random ensemble result for the same amplitudes [51]. Here the deterministic product interaction realizes that value through the zero set of its difference frequencies. The number N_{eff} measures channel participation in the prepared wavepacket. Equal weights maximize this participation and minimize the average purity at fixed channel count. Node locations determine the approach to that average and the recurrence pattern; the weights determine its value. The inset verifies this separation. Thus a register's Hilbert-space dimension and its effective participation play different roles in entanglement dynamics. Appendix E derives the averaging law and the continuum reference.

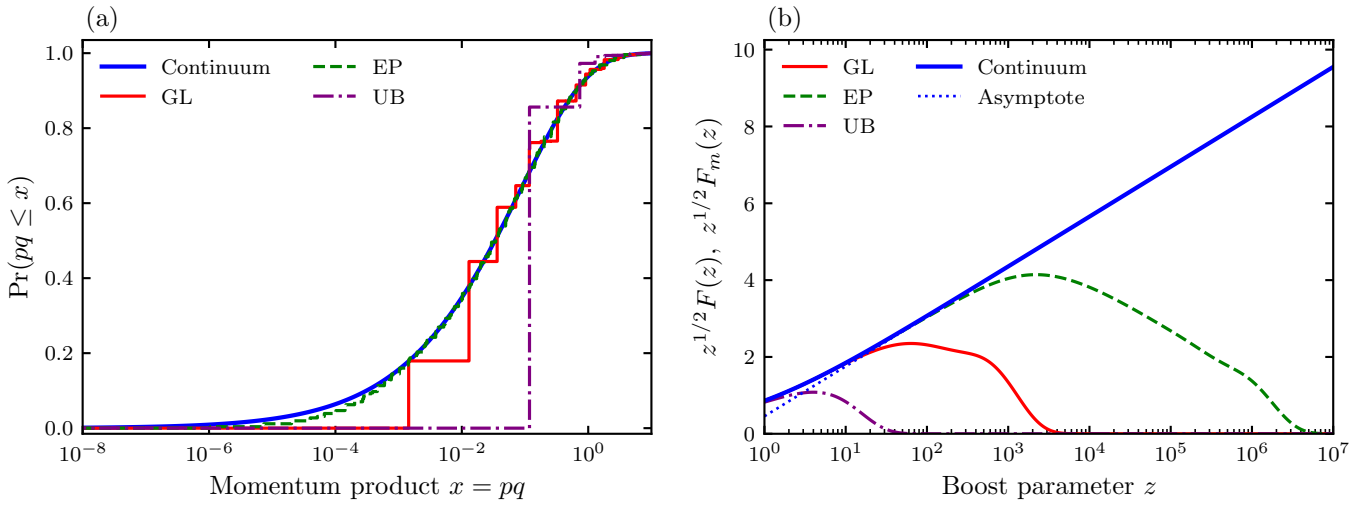


FIG. 3. Soft spectral weight and the regulated long-boost response, using the same $r = 1/2$, $m = 16$ measures as Fig. 2. (a) Cumulative probability of $x = pq$. Each discrete measure is drawn as a right-continuous staircase, starting at zero below its first product. (b) Rescaled correlator and the continuum asymptote. GL preserves exchange moments, EP resolves the small-momentum probability distribution, and UB uses uniform intervals on the common truncated support. The step positions and their masses determine the departures from the continuum curve.

V. SOFT MOMENTUM SELECTION BY THE HORIZON BOOST

The regulated response selects a different spectral variable. With $x = pq$, its density is

$$\rho_r(x) = \frac{2x^{r-1}}{\Gamma(r)^2} K_0(2\sqrt{x}), \qquad F(z) = \int_0^\infty \rho_r(x) e^{-zx}\, \mathrm{d}x. \tag{20}$$

Increasing the horizon boost suppresses large products and magnifies the contribution of soft pairs. Equal endpoint powers in the two wavepackets generate a logarithmic enhancement, giving

$$F(z) \sim \frac{z^{-r}}{\Gamma(r)}[\ln z - \psi(r) - 2\gamma_{\mathrm{E}}]. \tag{21}$$

The logarithm receives contributions from asymmetric pairs: one momentum can be soft while the other remains appreciable. The response therefore tests the distribution of low-product weight across scales.

Figure 3 shows how the different representations allocate weight to this sector. Quantile bins place more channels near the singular endpoint and retain the regulated response further into the boost evolution. Their advantage in this observable follows from the soft probability distribution. The unitary observables in Fig. 2 additionally sample phase differences throughout the spectrum, which explains their different convergence pattern. Agreement of the regulated response and fidelity of the real-parameter dynamics consequently require separate spectral tests.

The smallest node of the Gauss projection decreases inversely with chain length. For quantile bins its decrease is governed by the endpoint exponent. These different endpoint scales explain how the same channel count can resolve different soft regions. The first loss of response accuracy is determined by their full cumulative weights, whereas the smallest product controls the ultimate exponential tail of each finite representation. Appendix D derives the endpoint scales and their relation to the logarithmic continuum tail.

VI. EFFECTIVE WINDOWS, RESOURCES, AND REVERSAL

Following the observable-error criteria used in finite dynamical representations [43, 44, 46, 49], we define an effective window as the connected evolution interval from the initial point on which the chosen observable remains within its prescribed error tolerance. Figure 4(a) applies this criterion to the regulated correlator and expresses the result in boundary time. Figure 4(b) gives the encoding cost for a target response. The method dependence follows from how the available channels are placed and weighted. The Krylov boundary-feedback equation describes the complementary real-time effect of the conformal-chain cutoff. Each window is tied to its specified generator, observable, and accuracy criterion.

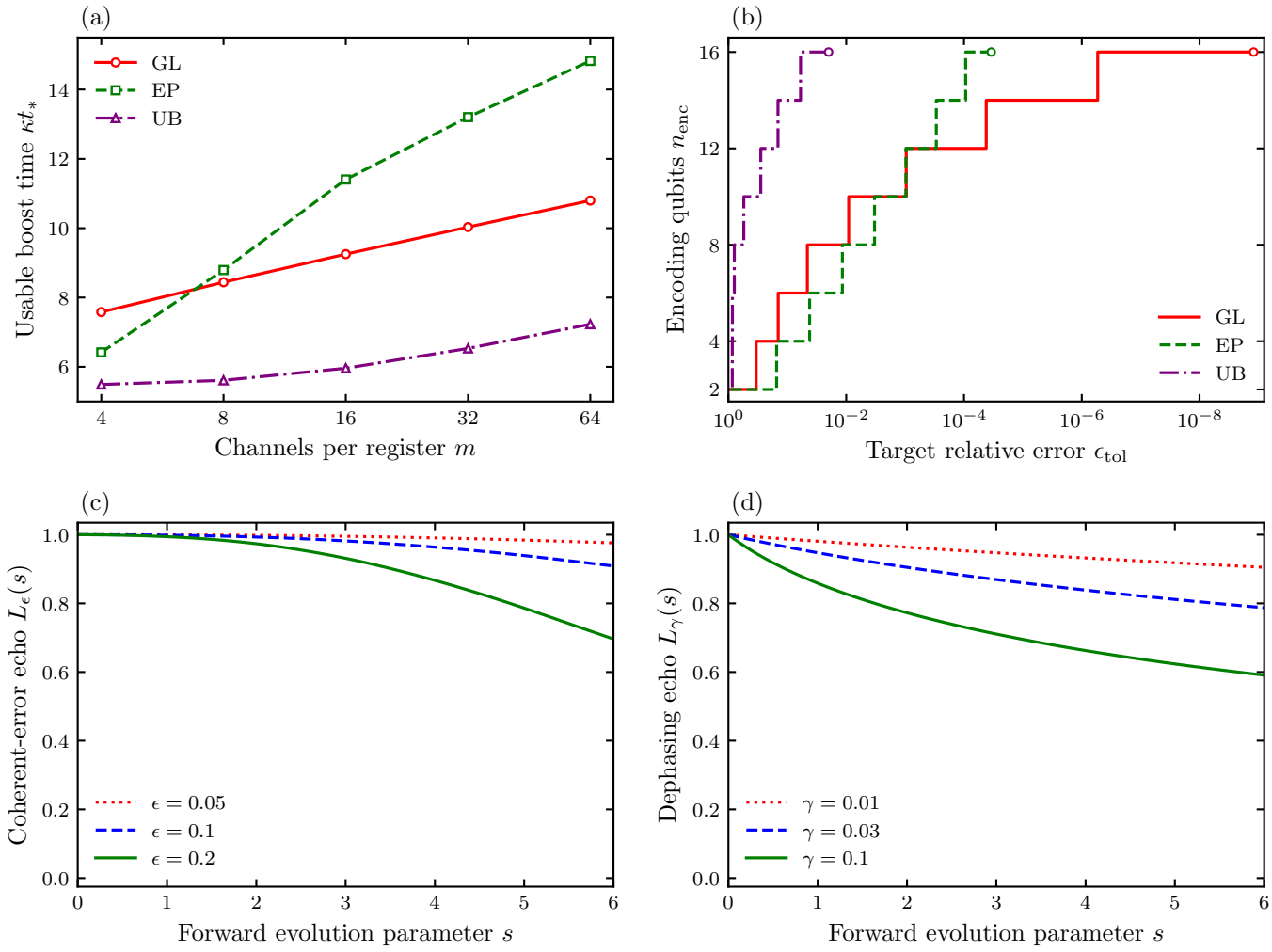


FIG. 4. Observable resolution and reversal robustness. (a) First 1% relative-error crossing of the regulated response, expressed as $\kappa t_* = \ln(z_*/z_0)$, with $z_0 = 10^{-3}$, $r = 1/2$, and $m = 4, 8, 16, 32, 64$. (b) Data-register cost $n_{\rm enc} = 2\log_2 m$, minimized over $m = 2, 4, \ldots, 256$ for the interval $0 \le z \le 10$. Curves cover the tolerances reached by the sampled resources. (c) Noncommuting coherent-error echoes for $\epsilon = 0.05, 0.1, 0.2$. (d) Local dephasing echoes for $\gamma = 0.01, 0.03, 0.1$. The two echo panels use the GL representation with $m = 16$, $r = 1/2$; each forward and reverse leg has duration s. The error operators are given in Appendix F.

The same register realization makes reversal errors explicit. A noncommuting perturbation to the reverse Hamiltonian transfers amplitude between momentum channels; dephasing suppresses coherence between their eigenstates. The echoes in Figs. 4(c) and 4(d) resolve these distinct mechanisms. Coherent-error loss begins quadratically in the evolution parameter, while Markovian dephasing produces linear initial attenuation. The response window describes spectral resolution, and the echo describes control of the prepared quantum state. Both enter the choice of a usable simulator.

This distinction also clarifies the quantum information retained by the projection. The channel measure retains specified correlation functions, and the joint phase produces correlations between the two registers. Preparation participation sets the infinite-time average of purity, while controlled reversal probes the preservation of the relative phases. These quantities provide explicit information diagnostics for the projected scattering sector.

VII. QUANTUM REALIZATION AND THE HOLOGRAPHIC CORRESPONDENCE

Figure 5 displays the correspondence between the boundary many-body system, near-horizon scattering, and the quantum toy model. The low-energy SYK probe fixes the conformal representation and its spectral measure. JT backreaction couples the two momentum sectors. Finite projection then produces a pair of quantum registers, each with a basis indexed by the retained channels. This sequence connects the operator-space dynamics of a strongly interacting system to a gravitational response and to a finite quantum evolution.

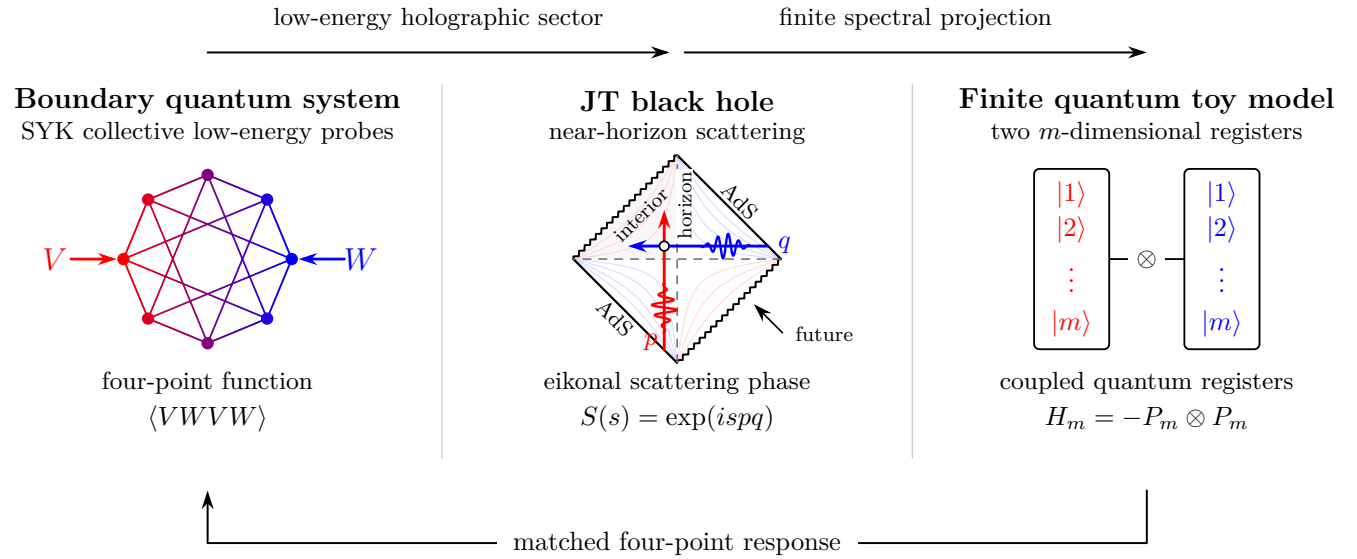


FIG. 5. From the SYK collective sector to a finite quantum realization of JT scattering. The boundary network uses a continuous red-to-blue gradient to connect the two probe labels. The central JT Penrose patch has a vertical horizon diagonal and an explicit future direction; red and blue wavepackets encode the two momentum measures. The right-hand registers retain the corresponding finite spectral channels and interact through H_m. The lower arrow denotes reconstruction of the designated four-point response from the weighted channel spectrum.

In a quantum implementation, each register is a controllable Hilbert space carrying the channel labels. Preparing the weights, implementing the conditional phase, and measuring the overlap or reduced purity realize the unitary observables. The regulated response is obtained from the same measured channel spectrum; equivalently, it is the success probability of the contraction $M_z = \exp[-zP_m \otimes P_m/2]$, since $F_m(z) = \|M_z|\psi_m\rangle\|^2$. The Krylov basis offers a complementary chain representation, with the momentum channels obtained by diagonalization. This basis freedom connects the construction to both collective many-body dynamics and programmable quantum systems.

VIII. CONCLUSIONS

Finite projections of the SYK conformal sector acquire a boundary term that directly controls the change in operator growth. The associated parabolic generator supplies the positive momentum channels of JT scattering, so the same projection also has a near-horizon interpretation through its soft spectral resolution. Real-parameter evolution probes momentum-difference interference and reveals a distinct participation-controlled average purity. These results identify the physical content retained by the finite Hamiltonian: a hierarchy of exchange amplitudes, a resolved range of horizon momenta, and a specified distribution of quantum coherence. Equal-resource comparisons show how these requirements select different channel constructions. The resulting criteria connect condensed-matter operator dynamics, holographic response, and quantum-register observables in a common spectral framework.

ACKNOWLEDGMENTS

The OpenQuantum plugin (https://github.com/xi-zhao/OpenQuantum), operating within Codex, led the theoretical derivations and code development in this work. The work was guided by the following prompt: "For the SYK model, what new quantum-physical insights are obtained by applying Gaussian quadrature to a known scattering kernel and expressing the result as a Hamiltonian? Relate fluctuations and dissipation in the SYK model as an open quantum system to black-hole thermodynamics, investigate the black hole's information properties, and map the resulting insights back to quantum information theory."

Appendix A: Conformal representation and boundary feedback

The normalized Wightman function in Eq. (3) fixes the Jacobi coefficients of its thermal spectral measure. The $\mathfrak{sl}(2,\mathbb{R})$ representation in Eq. (5) gives $\mathcal{L} = 2\alpha K_x$, with $K_x = (K_+ + K_-)/2$ and $K_y = (K_+ - K_-)/(2\mathrm{i})$. The infinite-chain probabilities are

$$|\varphi_n(t)|^2 = \frac{(r)_n}{n!}\,\mathrm{sech}^{2r}(\alpha t)\tanh^{2n}(\alpha t). \tag{A1}$$

Their mean is $r\sinh^2(\alpha t)$, and the weight beyond the retained space is $I_{\tanh^2(\alpha t)}(m, r)$. At fixed tail weight η,

its large-m crossing satisfies

$$\kappa t_{K,\eta} = \ln(4m/y_\eta) + O(m^{-1}), \qquad \frac{\Gamma(r, y_\eta)}{\Gamma(r)} = \eta. \quad \text{(A2)}$$

This is a tail-occupation criterion for the infinite conformal chain.

The null generators $P_\pm = 2(K_0 \pm K_y)$ obey

$$\mathrm{i}[\mathcal{L}, P_\pm] = \pm\kappa P_\pm. \quad \text{(A3)}$$

A rotation $R = \exp(-\mathrm{i}\pi K_0/2)$ gives $RPR^\dagger = P_-$. Since $R|0\rangle$ differs from $|0\rangle$ by a phase, the vacuum spectral measures coincide. The real tridiagonal P used for quadrature is this rotated positive generator.

Its orthonormal polynomials can be chosen as

$$\ell_n(p) = \sqrt{\frac{n!}{(r)_n}} L_n^{(r-1)}(p). \quad \text{(A4)}$$

Their multiplication recurrence has diagonal $2n + r$ and off-diagonal $-\sqrt{(n+1)(n+r)}$, with measure $\mathrm{d}\mu_r(p)$. The finite Jacobi matrix consequently has the Gauss nodes as eigenvalues and the squared first components of its normalized eigenvectors as weights. This gives Eq. (12) by a unitary basis change.

For the finite projection, the commutators with K_0 retain their form, while

$$[K_{-,m}, K_{+,m}] = 2K_{0,m} - D_m, \quad \text{(A5)}$$

$$D_m = m(m + r - 1)|m - 1\rangle\langle m - 1|. \quad \text{(A6)}$$

Thus $\dot{K}_0 = \kappa K_y$ and $\dot{K}_y = \kappa(K_0 - D_m/2)$. Taking expectations and subtracting the vacuum offset gives Eq. (8). The source is localized at the terminal site, with an amplitude fixed by the missing conformal matrix element.

Appendix B: Continuous scattering kernel

The equally regulated eikonal expression of Ref. [6] takes the form in Eq. (9) after rescaling the longitudinal momenta. In the convention $\beta = 2\pi$, the boost variable is $z = e^{\hat{u}}/(8C)$. The eikonal limit keeps $e^{\hat{u}}/C$ fixed as $C, \hat{u} \to \infty$. Integrating over one wavepacket gives

$$F(z) = \frac{1}{\Gamma(r)} \int_0^\infty p^{r-1} e^{-p} (1 + zp)^{-r} \, \mathrm{d}p = z^{-r} U(r, 1, z^{-1}). \quad \text{(B1)}$$

The small-z coefficients are $(-1)^n (r)_n^2/n!$. They define a right-sided asymptotic expansion; the required finite-order derivatives follow directly from the finite moments of the Gamma measure. For independent reference checks, the cases used below admit

$$F_{r=1/2}(z) = \frac{e^{1/(2z)} K_0(1/(2z))}{\sqrt{\pi z}}, \quad \text{(B2)}$$

$$F_{r=1}(z) = \frac{e^{1/z} E_1(1/z)}{z}. \quad \text{(B3)}$$

The exponentially scaled Bessel function avoids overflow in the first expression.

The distinction between a two-wavepacket kernel and a fixed source can also be expressed directly. Setting $q = q_0$ before the second average yields $(1 + zq_0)^{-r}$. This single-vertex form is related to the Gamma-vertex parametrization used in the scramblon analysis of Ref. [40]. Equation (9) retains both fluctuating wavepackets; the register construction performs positive discretization on each measure.

Appendix C: Moment preservation and a global error bound

Let I denote integration against the normalized Gamma measure and Q_m its m-point Gauss rule. Write

$$\mu_n = I[p^n] = (r)_n, \qquad \widehat{\mu}_n = Q_m[p^n].$$

The monic orthogonal polynomial is $\pi_m(p) = (-1)^m m! L_m^{(r-1)}(p)$, whose squared norm is

$$h_m = I[\pi_m^2] = m!(r)_m. \quad \text{(C1)}$$

The polynomial $p^{2m} - \pi_m(p)^2$ has degree at most $2m-1$, and π_m vanishes at every quadrature node. Exactness therefore gives

$$\mu_{2m} - \widehat{\mu}_{2m} = h_m. \quad \text{(C2)}$$

For fixed $q \geq 0$, let $H_q(p)$ be the Hermite interpolant of e^{-zpq}, matching its value and first derivative at every quadrature node. The pointwise remainder is nonnegative and satisfies

$$0 \leq e^{-zpq} - H_q(p) \leq \frac{(zq)^{2m}}{(2m)!} \pi_m(p)^2. \quad \text{(C3)}$$

This pointwise inequality is integrated at fixed q on the unbounded domain. Since $I[H_q] = Q_m[H_q] = Q_m[e^{-zpq}]$,

$$0 \leq (I_p - Q_p) e^{-zpq} \leq \frac{h_m z^{2m} q^{2m}}{(2m)!}. \quad \text{(C4)}$$

Applying the positive tensor-product decomposition

$$I_p I_q - Q_p Q_q = I_q(I_p - Q_p) + Q_p(I_q - Q_q)$$

then proves Eq. (16) with

$$c_m = \frac{h_m(\mu_{2m} + \widehat{\mu}_{2m})}{(2m)!} = \frac{h_m[2(r)_{2m} - h_m]}{(2m)!}. \quad \text{(C5)}$$

The first unmatched derivative gives

$$F(z) - F_m(z) = c_m z^{2m} + O(z^{2m+1}), \quad \text{(C6)}$$

so the coefficient in the global absolute-error bound is sharp at the origin. The same exactness argument proves Eq. (15).

Appendix D: Product spectrum and infrared response

Changing variables from (p, q) to $(p, x = pq)$ gives

$$\rho_r(x) = \frac{x^{r-1}}{\Gamma(r)^2} \int_0^\infty \frac{\mathrm{d}p}{p}\, e^{-p-x/p} \tag{D1}$$

$$= \frac{2x^{r-1}}{\Gamma(r)^2} K_0(2\sqrt{x}). \tag{D2}$$

The endpoint behavior is

$$\rho_r(x) = \frac{x^{r-1}}{\Gamma(r)^2} \left[-\ln x - 2\gamma_{\mathrm{E}} + O(x|\ln x|)\right]. \tag{D3}$$

Laplace transformation produces Eq. (21), with the remainder $O(z^{-r-1}\ln z)$. This form also follows from the small-argument expansion of $U(r, 1, z^{-1})$.

The discrete spectral measure is

$$\rho_m(x) = \sum_{j,k} w_j w_k \delta(x - p_j p_k). \tag{D4}$$

For ordered nodes, its ultimate large-boost response is $F_m(z) \sim w_1^2 e^{-zp_1^2}$. In boundary time this becomes $w_1^2 \exp[-z_0 p_1^2 e^{\kappa t}]$, whereas the continuous tail is proportional to $e^{-r\kappa t}$ times a term linear in t. The finite-window accuracy is determined before this ultimate limit by the full low-product spectral weight. In the single-integral form of Eq. (B1), the region $z^{-1} \ll p \ll 1$ contributes approximately $z^{-r}\mathrm{d}p/[\Gamma(r)p]$. Its logarithmic extent explains the role of asymmetric soft momentum pairs.

For the Gauss rule, the hard-edge asymptote of the first Laguerre zero gives

$$p_{\min}^{\mathrm{GL}} \sim \frac{j_{r-1,1}^2}{4m + 2r}, \tag{D5}$$

where $j_{\nu,1}$ is the first positive zero of J_ν. Equal-probability bins have first boundary $a_m \sim [\Gamma(r+1)/m]^{1/r}$. Its conditional mean is

$$p_{\min}^{\mathrm{EP}} \sim \frac{r}{r+1}\left[\frac{\Gamma(r+1)}{m}\right]^{1/r}. \tag{D6}$$

At a fixed omitted tail mass δ, the EP expression acquires a factor $(1-\delta)^{1/r}$. These asymptotes specify endpoint resolution. The finite-tolerance response windows are obtained from the complete spectral sums, including the asymmetric pairs responsible for the logarithmic tail.

Appendix E: Return probability and entanglement

The evolved register wavefunction is

$$\psi_{jk}(s) = \sqrt{w_j w_k}\, e^{\mathrm{i}sp_j p_k}. \tag{E1}$$

Tracing out one register gives

$$[\rho_P(s)]_{jj'} = \sqrt{w_j w_{j'}} \sum_k w_k e^{\mathrm{i}s(p_j - p_{j'})p_k}. \tag{E2}$$

Consequently,

$$\mathcal{P}_m(s) = \mathrm{tr}\,\rho_P(s)^2 \tag{E3}$$

$$= \sum_{j,j'} w_j w_{j'} \left|\sum_k w_k e^{\mathrm{i}s(p_j - p_{j'})p_k}\right|^2,$$

$$S_{2,m}(s) = -\ln \mathcal{P}_m(s). \tag{E4}$$

The continuous overlap is

$$A(s) = \int_0^\infty \mathrm{d}\mu_r(p)\,(1 - \mathrm{i}sp)^{-r}.$$

For the continuous purity, the second register can be integrated first:

$$\mathcal{P}(s) = \mathbb{E}_{p,p'}[1 + s^2(p-p')^2]^{-r}. \tag{E5}$$

At $r = 1/2$, the difference $d = p - p'$ has density $K_0(|d|)/\pi$, giving the independently evaluated reference

$$\mathcal{P}(s) = \frac{2}{\pi}\int_0^\infty \frac{K_0(d)}{\sqrt{1 + s^2 d^2}}\,\mathrm{d}d. \tag{E6}$$

The continuous integrals and finite pure-state partial traces are computed independently.

The continuum return probability and purity obey an additional identity at $r = 1/2$. Setting $a = (2s)^{-1}$ for $s > 0$, the Bessel representation of the amplitude and Nicholson's integral give

$$|A(s)|^2 = \frac{|K_0(\mathrm{i}a)|^2}{\pi s} = \frac{\pi}{4s}\left[J_0(a)^2 + Y_0(a)^2\right], \tag{E7}$$

$$\mathcal{P}(s) = \frac{2}{\pi s}\int_0^\infty K_0(s^{-1}\sinh u)\,\mathrm{d}u = |A(s)|^2. \tag{E8}$$

The second expression follows from $d = \sinh u/s$ in Eq. (E6). It gives the entropy–return relation used in the main text.

For any distinct finite nodes, expanding the purity gives

$$\mathcal{P}_m(s) = \sum_{j,l,k,n} w_j w_l w_k w_n e^{\mathrm{i}s(p_j - p_l)(p_k - p_n)}. \tag{E9}$$

All nonzero frequencies vanish under the infinite-time Cesàro average. The zero-frequency set is the union $j = l$ or $k = n$, with intersection counted once, proving Eq. (19). At finite averaging time T, the exact expression is

$$\overline{\mathcal{P}_m}^{(T)} = \sum_{j,l,k,n} w_j w_l w_k w_n\, \mathrm{sinc}[T(p_j - p_l)(p_k - p_n)], \tag{E10}$$

$$\mathrm{sinc}\, x = \sin(x)/x. \tag{E11}$$

This form evaluates the inset without time-discretization error. The entropy plotted in the main panel is the instantaneous $-\ln \mathcal{P}_m(s)$. Convexity separately gives the time-average inequality $\overline{S_{2,m}} \geq -\ln \overline{\mathcal{P}_m}$. The latter quantity is an entropy bound constructed from average purity.

Appendix F: Coherent errors and dephasing

The coherent-error protocol uses forward evolution under H_m and backward evolution under $-H_m - \epsilon V$. With

$$T_m = \sum_{j=1}^{m-1} |j+1\rangle\langle j|, \qquad X_m = (T_m + T_m^\dagger)/2, \tag{F1}$$

$$V = X_m \otimes I + I \otimes X_m, \tag{F2}$$

the echo is

$$L_\epsilon(s) = \left| \langle \psi_m | e^{\mathrm{i}(H_m + \epsilon V)s} e^{-\mathrm{i}H_m s} | \psi_m \rangle \right|^2. \tag{F3}$$

The perturbation connects different momentum eigenchannels and therefore does not commute with H_m.

For dephasing, the jump operators on both legs are

$$L_P = \sqrt{\gamma}\, P_m \otimes I, \qquad L_Q = \sqrt{\gamma}\, I \otimes P_m. \tag{F4}$$

We use $\dot\rho = -\mathrm{i}[H,\rho] + \sum_a (L_a \rho L_a^\dagger - \{L_a^\dagger L_a, \rho\}/2)$, with $H = H_m$ on the forward leg and $H = -H_m$ on the reverse leg. Each leg has duration s. In the product eigenbasis, the Hamiltonian phases cancel and the two legs multiply off-diagonal elements by

$$\exp\{-\gamma s[(p_j - p_{j'})^2 + (p_k - p_{k'})^2]\}.$$

The echo factorizes:

$$L_\gamma(s) = \left[\sum_{j,j'} w_j w_{j'} e^{-\gamma s (p_j - p_{j'})^2} \right]^2. \tag{F5}$$

This formula and the full density matrix are checked against independent forward and reverse Lindblad evolution.

Expanding the two protocols at short times gives

$$L_\epsilon(s) = 1 - \epsilon^2 \operatorname{Var}_{\psi_m}(V) s^2 + O(s^3), \tag{F6}$$

$$L_\gamma(s) = 1 - 4\gamma r s + O(s^2), \qquad m \geq 2. \tag{F7}$$

The dephasing coefficient uses the preserved variance of the Gamma measure. These laws connect the different initial echo profiles to coherent amplitude redistribution and irreversible loss of channel coherence.

Appendix G: Microscopic SYK spectral construction

We construct Eq. (1) with Jordan–Wigner Majoranas, retaining both fermion-parity sectors. For a Hermitian probe O, the exact normalized Wightman spectral measure is

$$\mathrm{d}\nu_{N,\beta,O}(\omega) = \frac{1}{\mathcal{N}} \sum_{a,b} \sqrt{\rho_a \rho_b} |O_{ab}|^2 \delta[\omega - (E_a - E_b)]\, \mathrm{d}\omega, \tag{G1}$$

where $\rho_a = e^{-\beta E_a}/Z$ and $\mathcal{N}$ is the total weight. It is even in ω. Lanczos recursion acts on multiplication by ω, starting from the square roots of these weights. Two complete reorthogonalization passes stabilize the recurrence. Its tridiagonal matrix reproduces the microscopic spectral moments through its quadrature degree; the conformal coefficients in Eq. (3) provide the low-energy reference [10, 11]. The frequency measure ν and the parabolic momentum measure μ_r are associated with their respective generators.

The implementation benchmark uses $N = 12, 16$, two disorder samples for each size, $\beta J = 0, 2, 8$, and 61 time points over $0 \leq Jt \leq 15$. The random seed is $20261025 + 1000N + a$, with sample index $a = 0, 1$. The first 16 Lanczos coefficients are obtained from the Majorana Wightman measure. Exact thermal traces, spectral sums, and the Lanczos moments provide independent checks. At the lowest simulated temperature, the finite-size Wightman curves approach the conformal profile while their Lanczos sequences retain mode-dependent departures. These two diagnostics separately track the shape of the correlator and the operator hierarchy. The four-point routines retain the fermionic sign and use graded commutators for odd probes, following the microscopic observable conventions relevant to SYK chaos [8]. Complete couplings, spectra, and trajectories are retained with the source.

Appendix H: Numerical procedure and reproducibility

All calculations use the fixed OpenQuantum environment with Python 3.12.14, QuTiP 5.3.1, NumPy 2.5.3, and SciPy 1.18.1. Native QuTiP operators, matrix exponentials, partial traces, and master-equation evolution supply independent checks of the spectral formulas. The conformal-chain calculations diagonalize $\mathcal{L}_m/\kappa$ and P_m separately. The terminal feedback is evaluated both from occupation probabilities and from $-[\mathcal{L}_m, [\mathcal{L}_m, n]]/\kappa^2$. On the plotted grids their difference is below 10^{-14}; independent QuTiP evolution agrees with the mean Krylov positions to within 3×10^{-14}.

The GL rule uses the zeros of $L_m^{(r-1)}$ and normalized positive weights. For the bin constructions we set a common tail mass $\delta = 10^{-12}$, choose L from $\Pr(p > L) = \delta$, and use intervals $[a_j, a_{j+1}]$ with either equal width or equal Gamma probability. With g_r denoting the Gamma density,

$$w_j = \frac{\int_{a_j}^{a_{j+1}} g_r(p)\, \mathrm{d}p}{1 - \delta}, \qquad p_j = \frac{\int_{a_j}^{a_{j+1}} p\, g_r(p)\, \mathrm{d}p}{\int_{a_j}^{a_{j+1}} g_r(p)\, \mathrm{d}p}. \tag{H1}$$

These centroids preserve the mean of the truncated measure. The exact normalized truncation changes the two-copy Laplace response by at most $1-(1-\delta)^2$; discretization error is evaluated against the full Gamma target. Equal channel count fixes the data-register dimension. Preparation amplitudes, spectral bandwidths, and synthesis costs are recorded separately.

Continuous references use scaled Bessel functions and independent adaptive integrals. The substitution $p = x^{1/r}$ removes the endpoint power singularity. Reference checks use requested absolute and relative integration tolerances of 2×10^{-12}, including independent logarithmic-coordinate integrals. The $r = 1/2$ return and purity identity is checked against separate integrals. QuTiP partial traces and spectral purity sums agree to within 10^{-15} in the comparison checks.

The equal-channel scan covers $m = 4, 8, 16, 32, 64$. Thresholds are defined from the first sampled crossing and refined with Brent's method. Dense preceding-interval scans check the selected crossings; the tested relative root changes are below 2×10^{-11}. The resource calculation evaluates the maximum sampled error across the target boost interval and refines interior maxima. The code also records cutoff and tolerance sensitivity. The purity inset integrates each discrete frequency exactly with the sinc kernel, over averaging times up to 10^6. Coherent-error echoes are checked against direct matrix exponentials and dephasing echoes against forward and reverse Lindblad evolution. The microscopic benchmark's independent Wightman traces agree within 4×10^{-16}, and its Lanczos spectral moments through order eight agree within 4×10^{-15} relative error. Input conventions, complete arrays, seeds, dependency versions, source hashes, and the individual check results accompany the scripts.

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